

Lemma: $y = 2\sin(x + \gamma)$

max: 9

min: 5

$T = \frac{\pi}{8}$

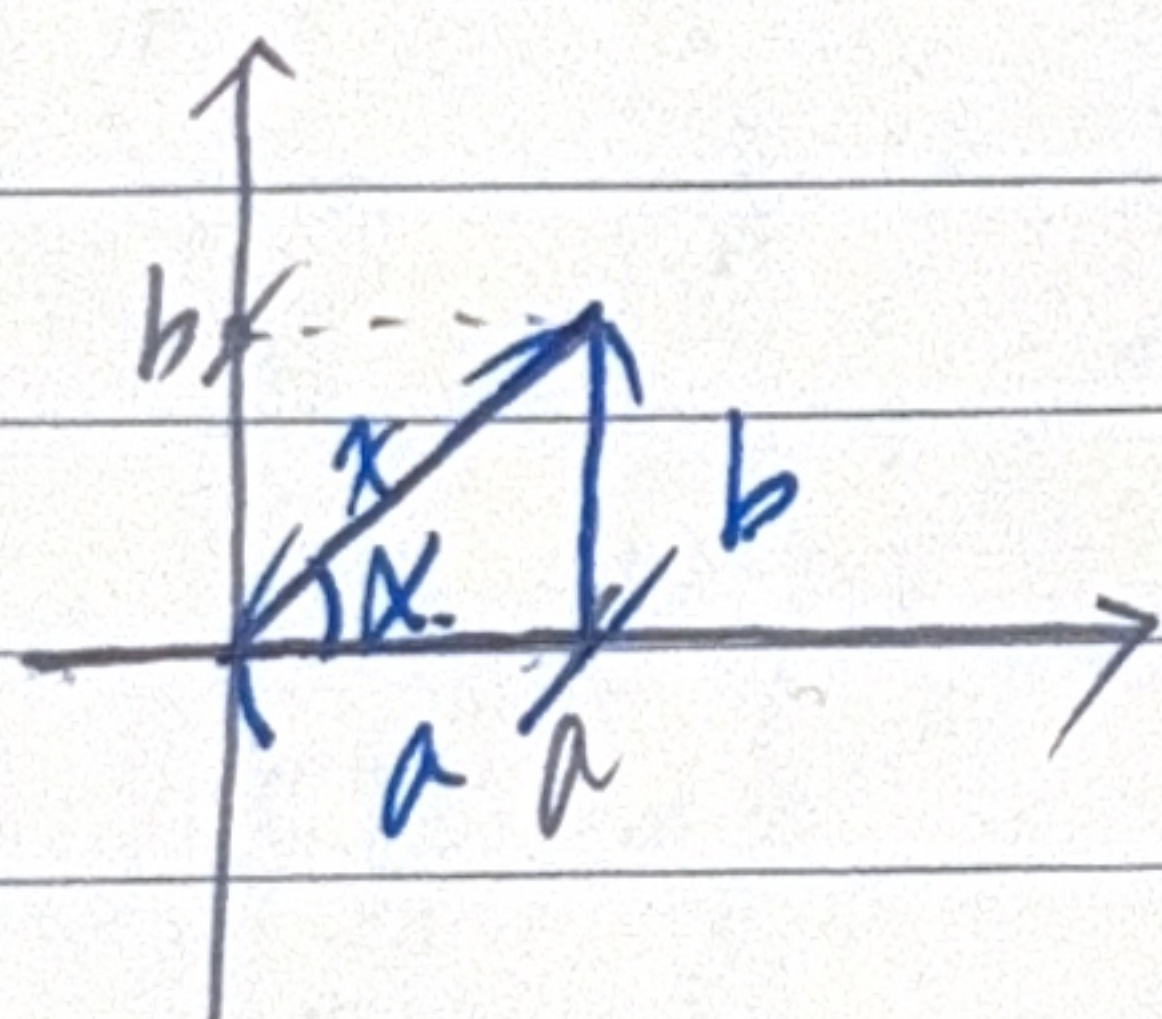
$\Rightarrow \sin(\theta + \alpha) \quad (-: \text{angle})$

$a\sin\theta + b\cos\theta = \sqrt{a^2+b^2}$

$\left(\sin\theta \cdot \frac{a}{\sqrt{a^2+b^2}} + \cos\theta \cdot \frac{b}{\sqrt{a^2+b^2}} \right)$

$\Rightarrow \cos\alpha$

$\Rightarrow \sin\alpha$



$r = \sqrt{a^2+b^2}$

$\sin\alpha = \frac{b}{\sqrt{a^2+b^2}}$

$\cos\alpha = \frac{a}{\sqrt{a^2+b^2}}$

$\Rightarrow f(x) = a\sin x + b\cos x$

$= \sqrt{a^2+b^2} \sin(x+\alpha)$

$\cdot M = \sqrt{a^2+b^2}$

$\cdot m = -\sqrt{a^2+b^2}$

$\cdot T = 2\pi$

